

ogy, it has been around for years. However, customer adoption has been low due to their slightly higher price. No doubt, there is a shift to sinewave inverters but that shift is still very slow.

Right now, the price difference between sinewave and squarewave inverters is very nominal at only around Rs 400.

What size of inverter do I need?

When buying an inverter, be clear about the number of appliances, bulbs and fans that you want to be run on the inverter in case of power failure. Buy an inverter that is of slightly higher capacity than you need at present as you will be able to run something else simultaneously, whenever required. So calculate the total power consumption requirement of the desired loads and go for slightly higher wattage.

Going by this, for a two-bedroom house, to run three lights, two fans and a TV set, about 600VA inverter would suffice. This is how the inverter capacity is arrived at in this case:

Power consumption of
Three 60W lights: 180W
Two ceiling fans: 150W
One 48.3cm (19-inch) colour TV: 100W

The total power requirement in this case being 430W, a 600VA inverter will suffice.

Buy a 5kVA inverter in case you also want to run an AC along with fans, tubelights and TV set.

Some appliances require extra load (called surge load) when these start up. While calculating the load requirement, take into account this surge load (startup load) and not the lower running load. Typically, the startup load



UTL sinewave combo UPS

is two times the running load. In the case of appliances like air-conditioners, refrigerators and pumps, it could be more than three times the continuous load rating.

How long can I run the appliances?

The battery capacity determines the backup duration. It is primarily defined in ampere hours (Ah) and selected on the basis of backup requirements of an individual. The more the Ah, the longer the backup. The number of plates and the total battery-plate surface area of the cells determine the length of backup. A 180Ah normal battery has 25 plates, while a 180Ah normal tubular battery has only 13 plates but greater plate area.

A standard battery provides three hours of backup time at full load and six hours at half load. You'll need a 180Ah battery for 800VA load.

You can increase the backup duration by installing a higher-capacity battery or adding extra batteries in parallel.

Standard, tubular or SMF?

Battery selection is very important as it's a recurring cost every 2-3 years. An ideal solution for inverters is either a thick-, flat-plate (standard) battery or a tubular battery. Both the thick-plate and tubular batteries need topping up. Of the two, tubular is better because it lasts longer. Flat batteries generally do not exceed three years of lifetime as against five years in the case of tubular batteries. But, the initial cost of tubular batteries is also higher.

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